Scattering versus accumulated exposure

Addendum to kapton_final_report_2026-10-26.pdf

DATA RECOVERED AND REDUCED

211 recorded image sets: 211 WAXS and 149 SAXS patterns reduced. Exposure-series figures use 201 WAXS and 140 SAXS beam-confirmed patterns.

Scan ID(s)	Acquired	Detectors	Accepted	Campaign h
1185420–1185481	Oct 22	WAXS	61	0.039–4.972
1185487	Oct 24	SAXS + WAXS	27	14.045–16.207
1185489	Oct 24	SAXS + WAXS	0	Dark only
1185490	Oct 24–26	SAXS + WAXS	113	16.333–25.621

HOW TO READ THIS ADDENDUM

Pages 2–6 show exposure-colored waterfalls and all-frame change maps. Page 7 quantifies within-session changes without vertical offsets. Page 8 checks masks and geometry; page 9 records methods and scan IDs.

The exposure coordinate is the campaign beam-on counter, including the documented +116.029 s early correction. Each session is compared to its own initial patterns. The first saved image already follows irradiation; it is not virgin Kapton. Friday’s approximately 8.88 h image-free irradiation interval cannot be reconstructed from scattering images.

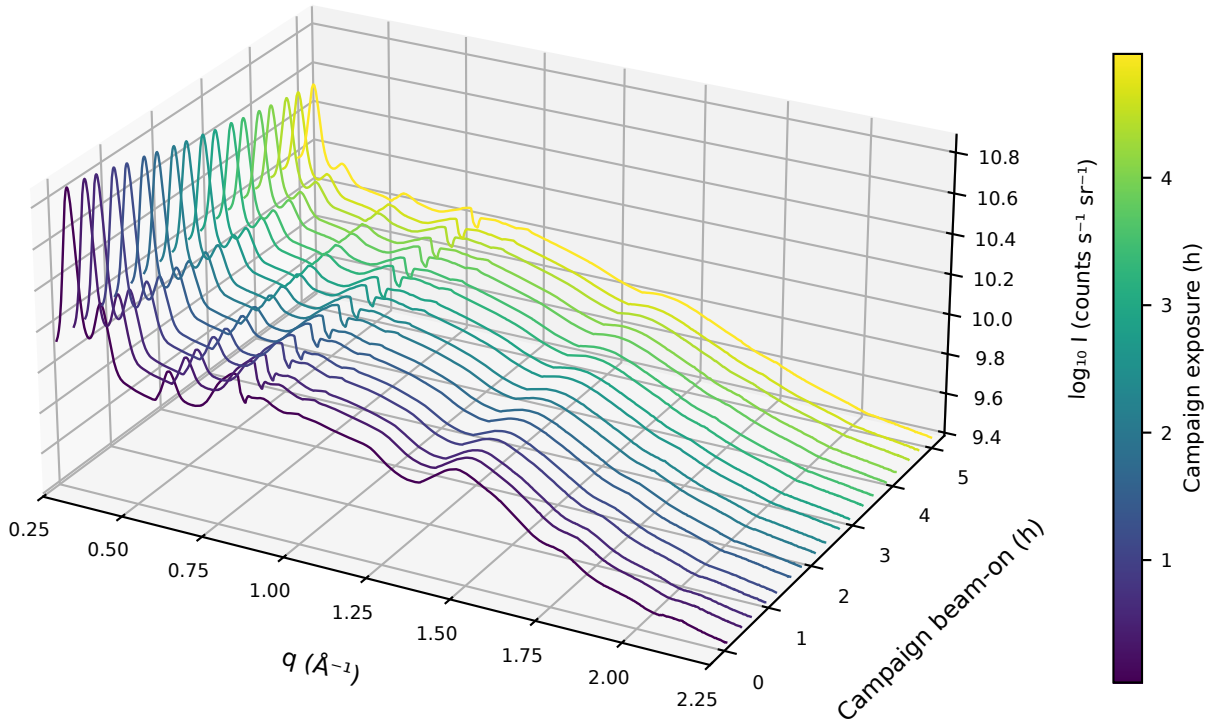
Eight post-trip frames in scan 1185487 are reduced for audit but excluded from exposure trends. Also excluded: 2 detector-dark sets (1185427 event 1, 1185489 event 1), despite pre-trigger beam-on readings. Scans 1185419, 1185486 and 1185488 contain no primary images. Scans 1185482–1185485 are beam-down acceptance tests. The final incomplete attempt in 1185490 has no primary image and is omitted.

LONG-SESSION ENDPOINT COMPARISON

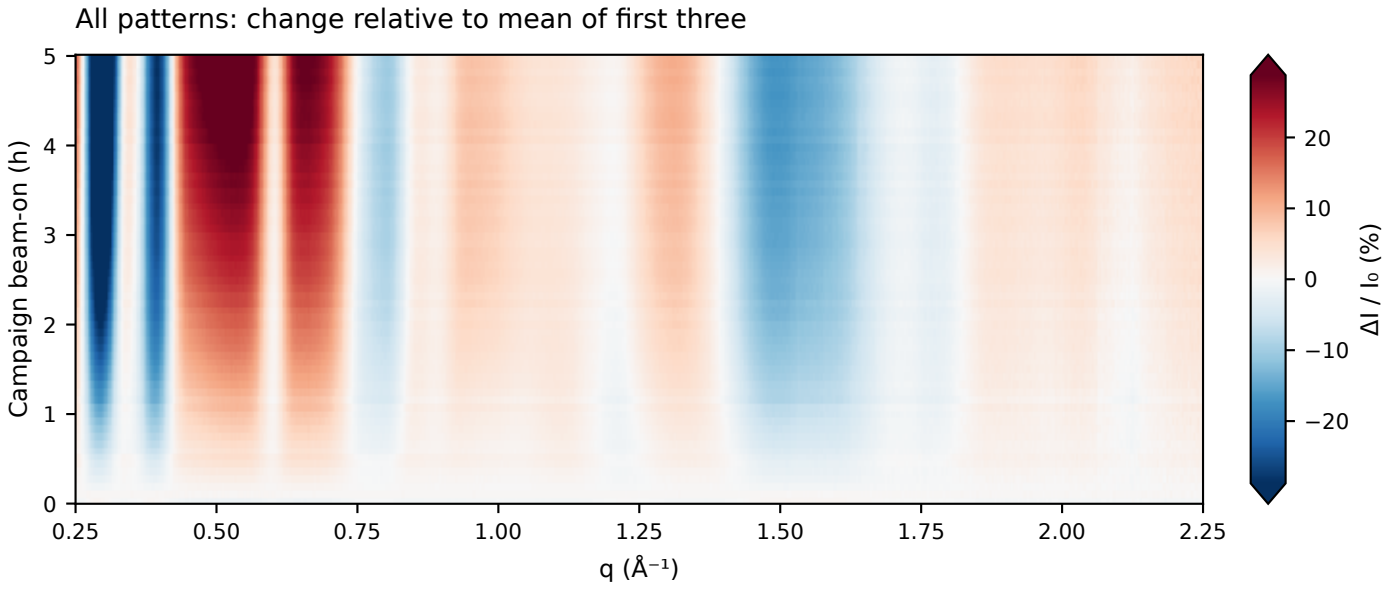
WAXS 0.4–0.8 $Å^{-1}$: +8.2%; WAXS 1–1.8 $Å^{-1}$: +2.7%; SAXS 0.002–0.008 $Å^{-1}$: +21.3%; SAXS 0.01–0.025 $Å^{-1}$: +23.7%; SAXS 0.03–0.06 $Å^{-1}$: +26.7%. Mean of last three versus first three, BPM3-normalized. Evolution is non-monotonic, with major steps across overnight beam gaps. The last three broad-WAXS values have 3.4% CV; a single first/last pair instead gives -0.72%. These are observed scattering changes, not an isolated material-damage measurement; sample/beam drift and transmission changes may contribute.

WAXS · exposure waterfall

Oct 22 · WAXS-only | 61 accepted patterns | 4.933 h beam-on between endpoints



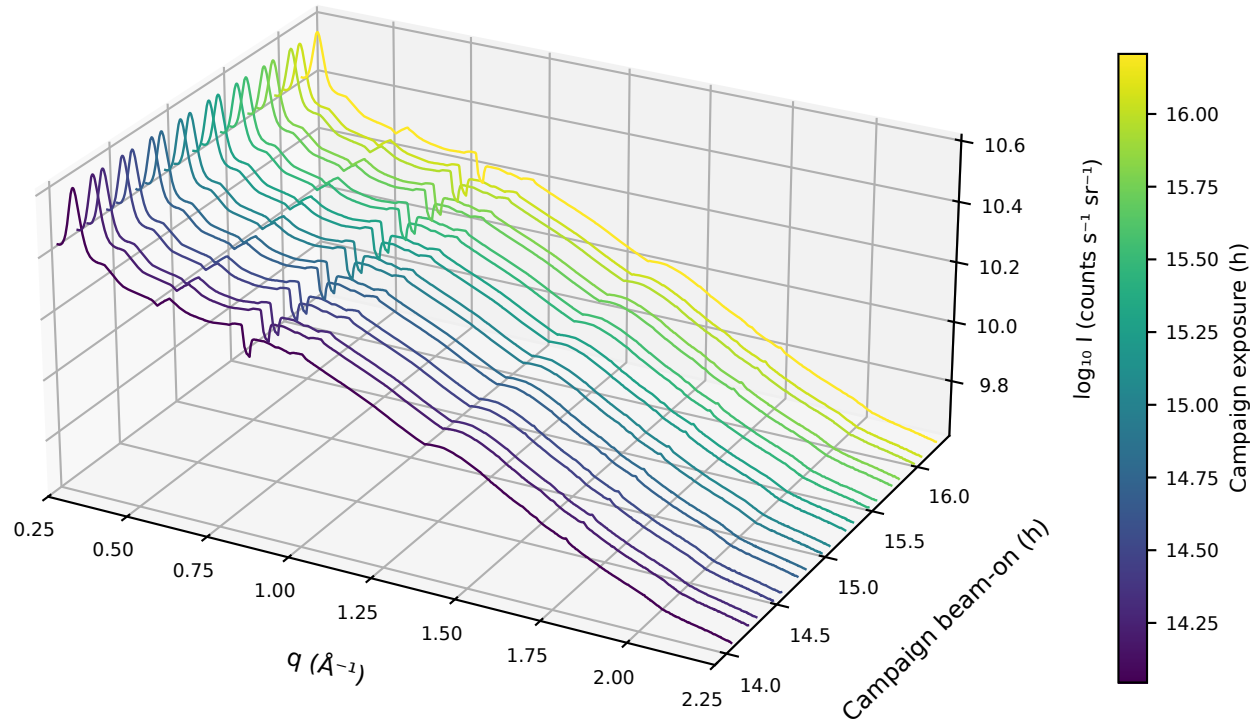
18 curves displayed, selected at approximately equal beam-time intervals.



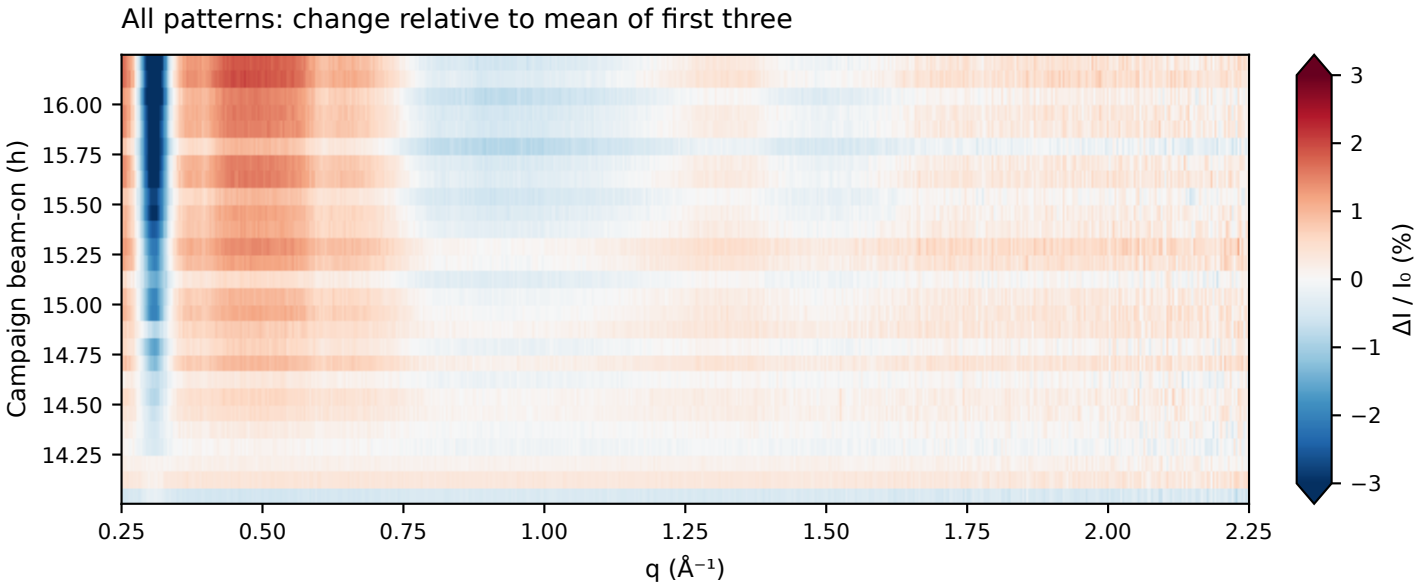
Intensity is divided by the detector exposure and pixel solid angle, then scaled to BPM3 = 50. Waterfall heights retain intensity differences; no per-curve area scaling is used. White q bins lack sufficient shared acceptance (at least 20 weighted pixels/bin). Heatmap rows use beam-time midpoints, not wall time; gaps in delivery add no exposure. The campaign counter includes earlier positions and is not a spot-specific dose.

WAXS · exposure waterfall

Oct 24 · morning session | 27 accepted patterns | 2.161 h beam-on between endpoints



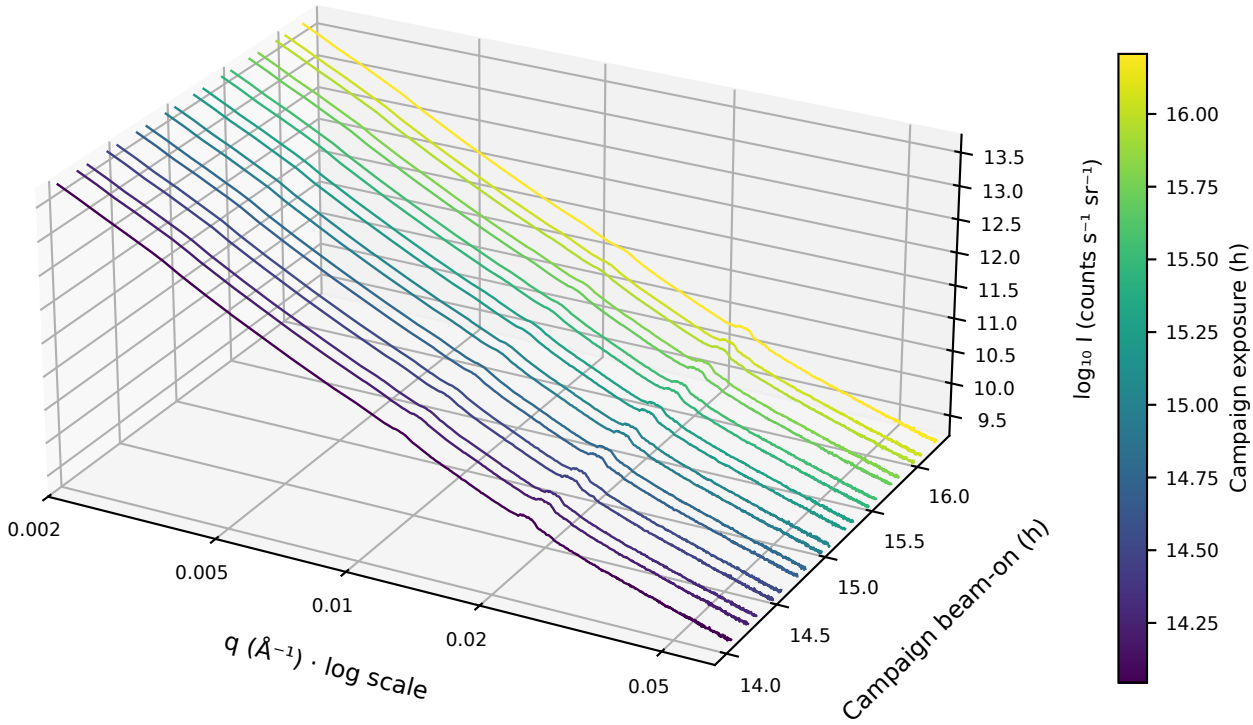
18 curves displayed, selected at approximately equal beam-time intervals.



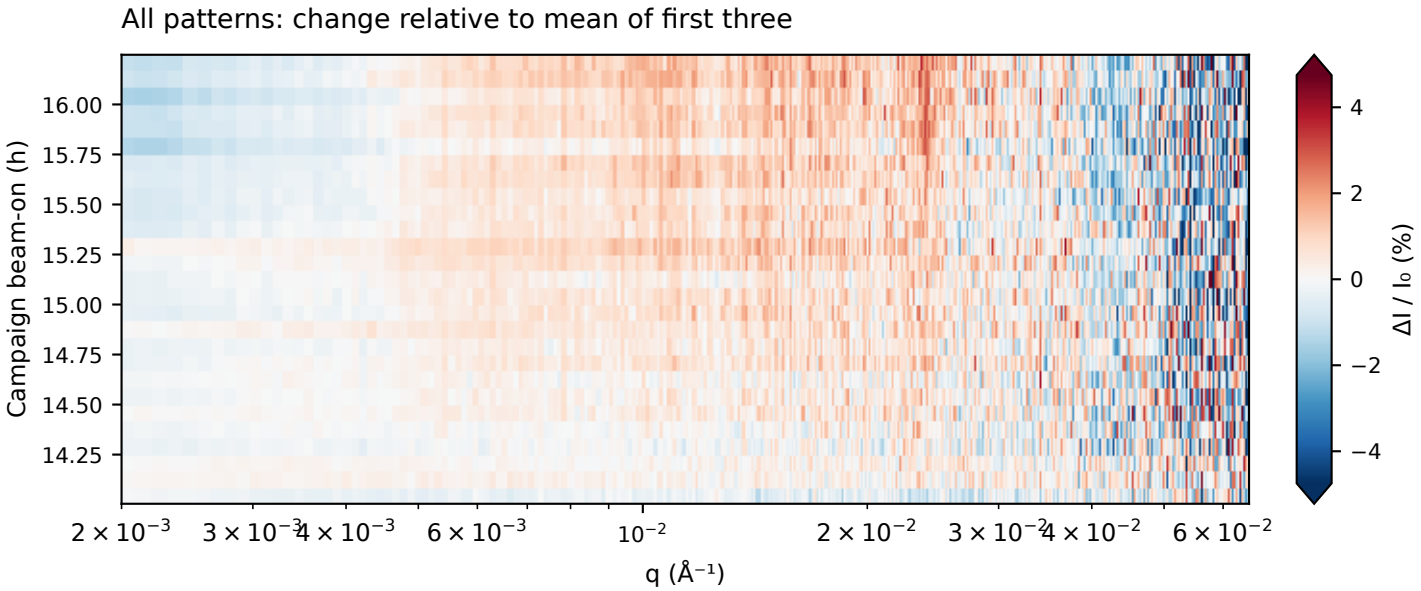
Intensity is divided by the detector exposure and pixel solid angle, then scaled to BPM3 = 50. Waterfall heights retain intensity differences; no per-curve area scaling is used. White q bins lack sufficient shared acceptance (at least 20 weighted pixels/bin). Heatmap rows use beam-time midpoints, not wall time; gaps in delivery add no exposure. The campaign counter includes earlier positions and is not a spot-specific dose.

SAXS · exposure waterfall

Oct 24 · morning session | 27 accepted patterns | 2.161 h beam-on between endpoints



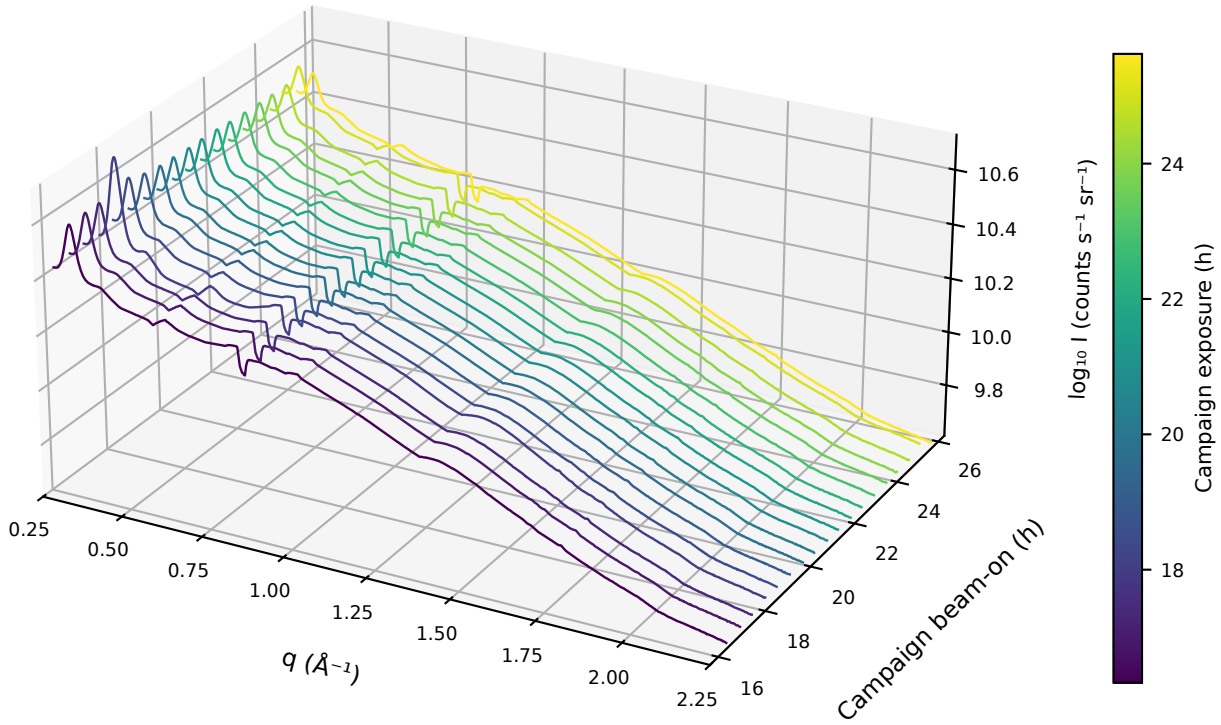
18 curves displayed, selected at approximately equal beam-time intervals.



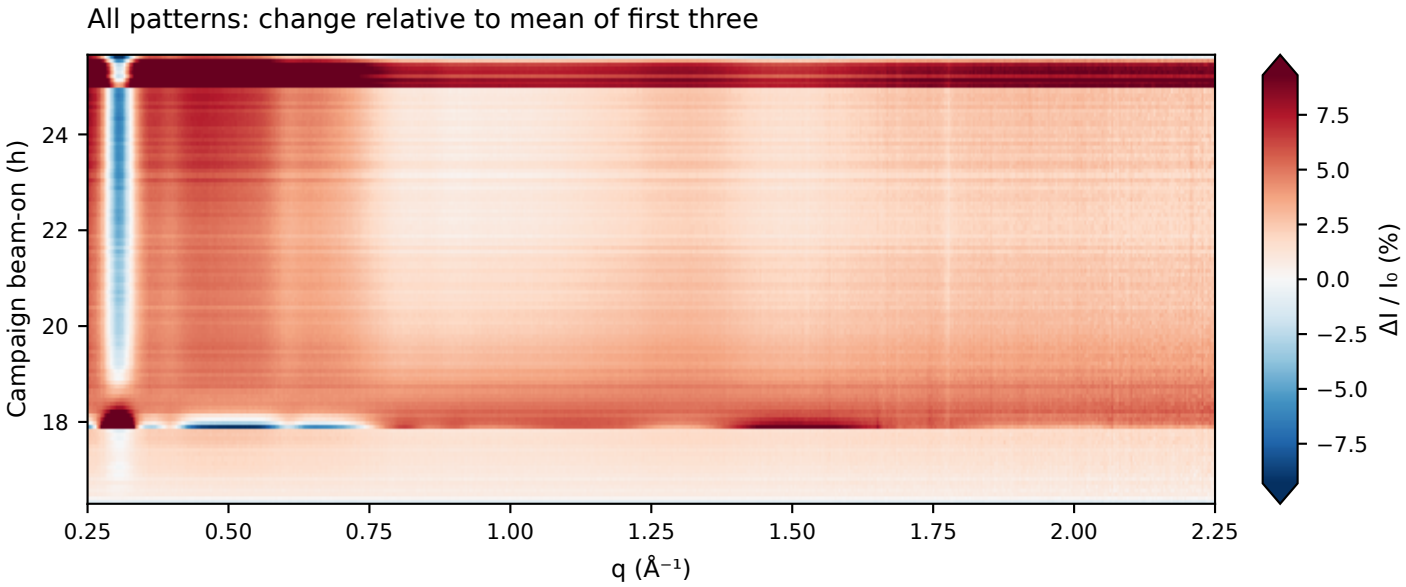
Intensity is divided by the detector exposure and pixel solid angle, then scaled to BPM3 = 50. Waterfall heights retain intensity differences; no per-curve area scaling is used. White q bins lack sufficient shared acceptance (at least 20 weighted pixels/bin). Heatmap rows use beam-time midpoints, not wall time; gaps in delivery add no exposure. The campaign counter includes earlier positions and is not a spot-specific dose.

WAXS · exposure waterfall

Oct 24–26 · long session | 113 accepted patterns | 9.288 h beam-on between endpoints



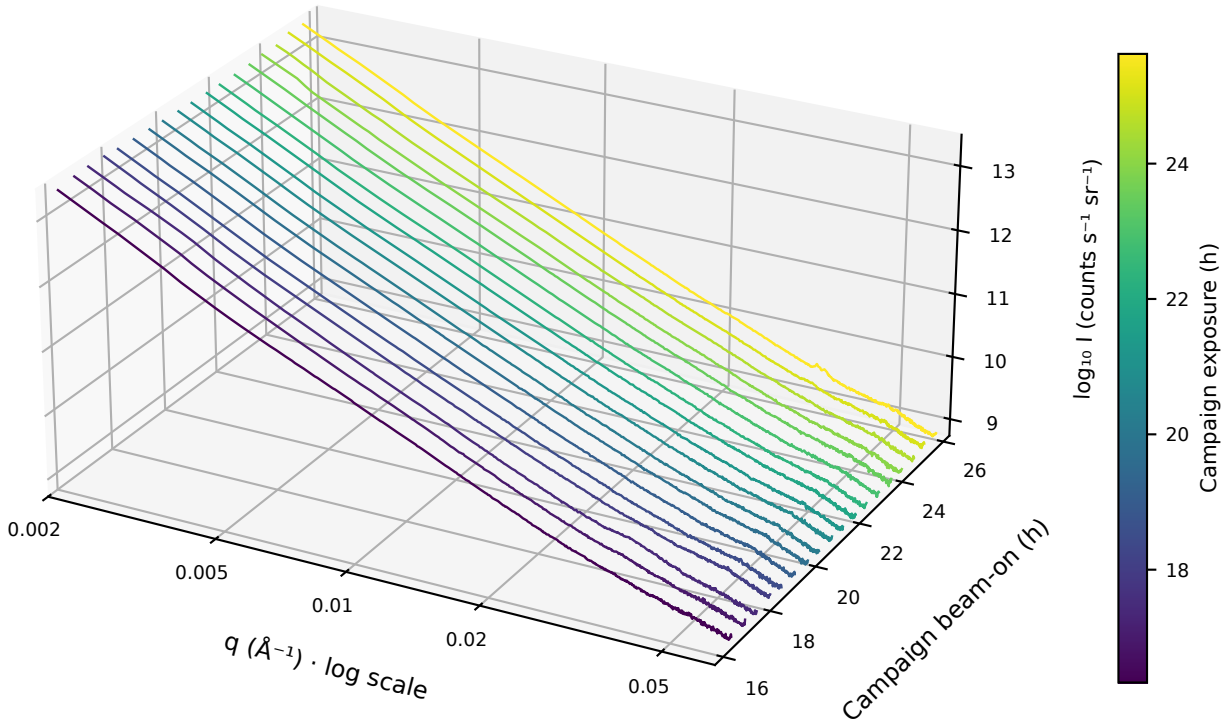
18 curves displayed, selected at approximately equal beam-time intervals.



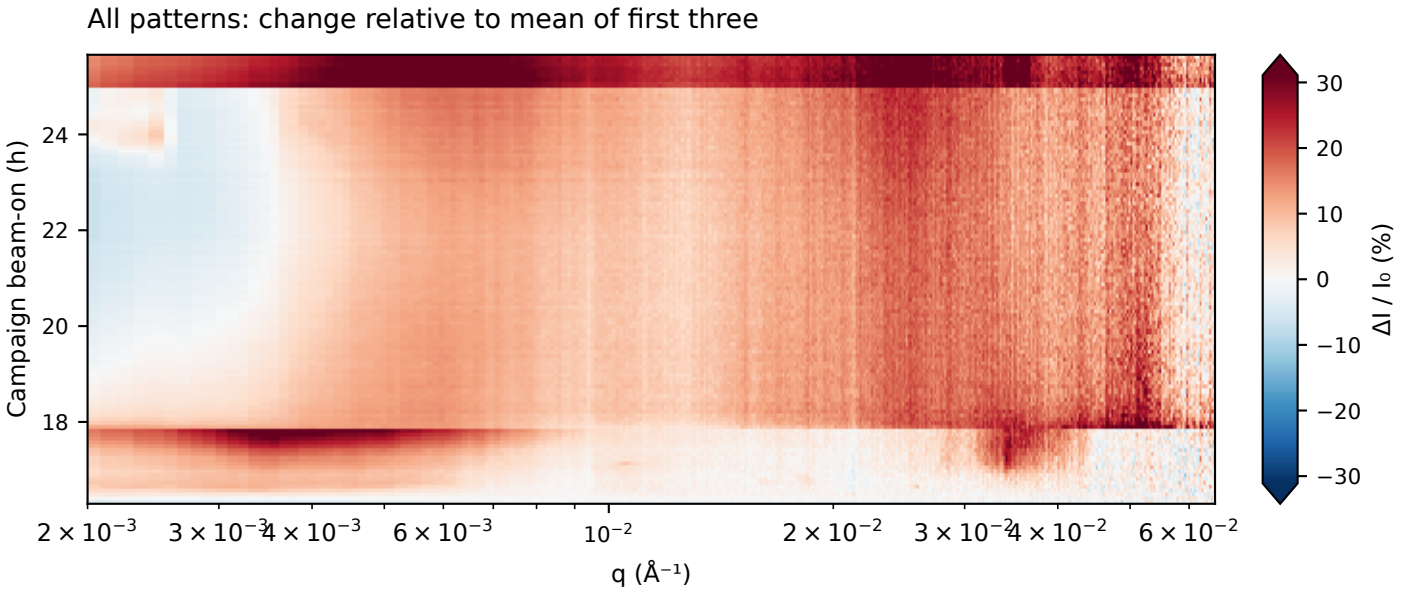
Intensity is divided by the detector exposure and pixel solid angle, then scaled to BPM3 = 50. Waterfall heights retain intensity differences; no per-curve area scaling is used. White q bins lack sufficient shared acceptance (at least 20 weighted pixels/bin). Heatmap rows use beam-time midpoints, not wall time; gaps in delivery add no exposure. The campaign counter includes earlier positions and is not a spot-specific dose.

SAXS · exposure waterfall

Oct 24–26 · long session | 113 accepted patterns | 9.288 h beam-on between endpoints



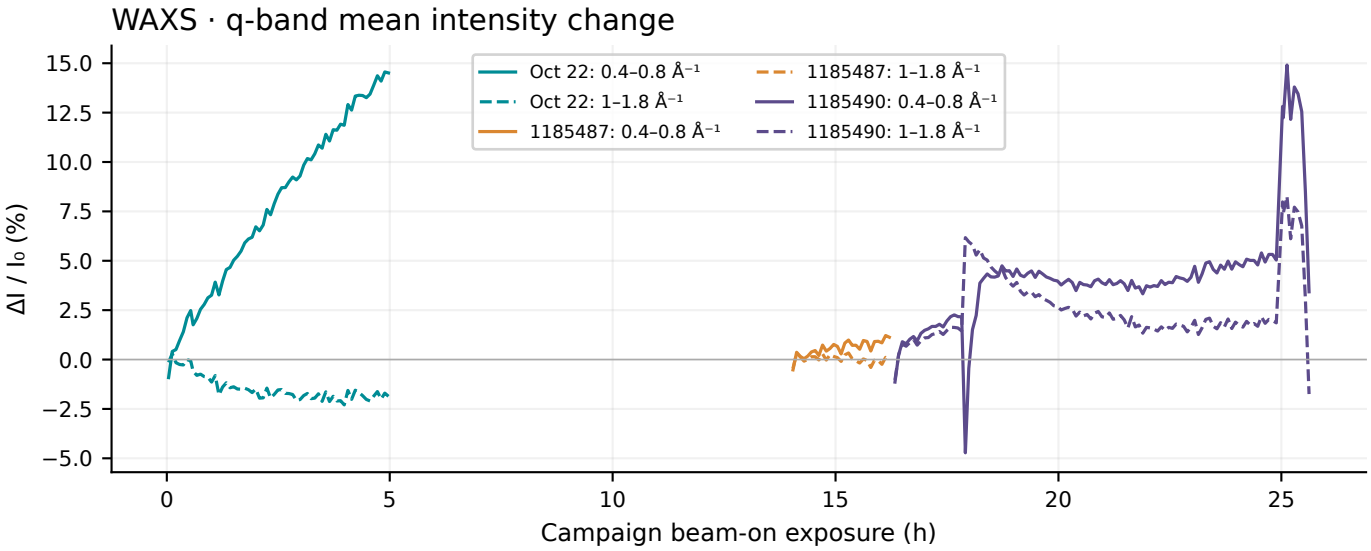
18 curves displayed, selected at approximately equal beam-time intervals.

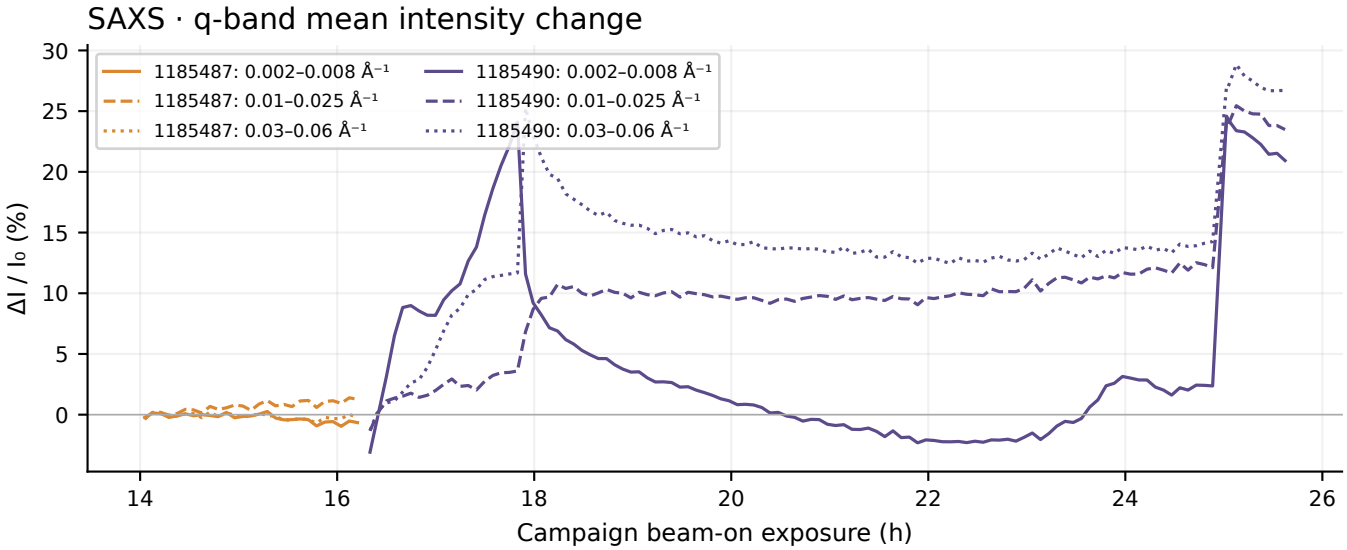


Intensity is divided by the detector exposure and pixel solid angle, then scaled to BPM3 = 50. Waterfall heights retain intensity differences; no per-curve area scaling is used. White q bins lack sufficient shared acceptance (at least 20 weighted pixels/bin). Heatmap rows use beam-time midpoints, not wall time; gaps in delivery add no exposure. The campaign counter includes earlier positions and is not a spot-specific dose.

Quantifying the scattering change

No waterfall offsets | Relative to each session’s first three accepted patterns





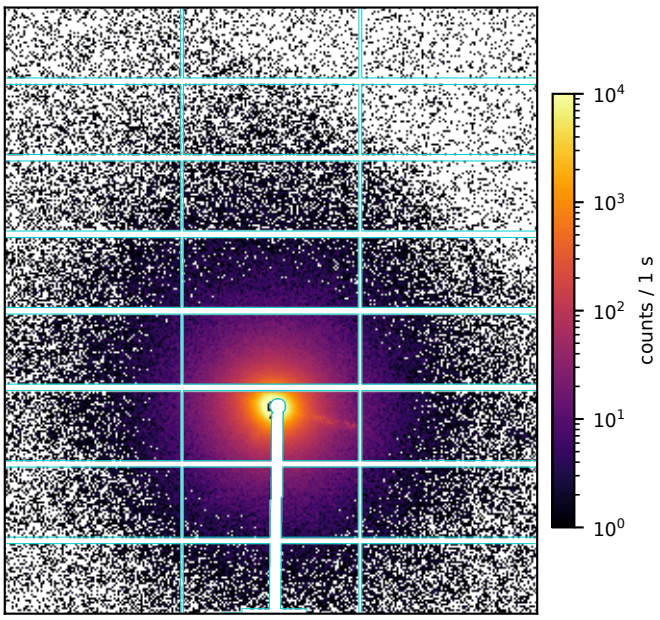
1185490	q (Å⁻¹)	BPM3 norm.	Time norm. only	CV first / last
WAXS	0.4–0.8	+8.16%	+9.74%	0.85% / 3.45%
WAXS	1–1.8	+2.69%	+4.19%	0.76% / 3.40%
SAXS	0.002–0.008	+21.30%	+23.08%	2.53% / 0.22%
SAXS	0.01–0.025	+23.70%	+25.51%	1.01% / 0.13%
SAXS	0.03–0.06	+26.69%	+28.54%	0.96% / 0.01%

Large steps bracket overnight beam gaps (1185490 events 19→20 and 104→105). Endpoints average three frames each. CV is within-window scatter, not total measurement uncertainty. Curves are joined only within a session; q-band means use bins valid in every included frame. No background, transmission, polarization, or absolute-intensity calibration is applied.

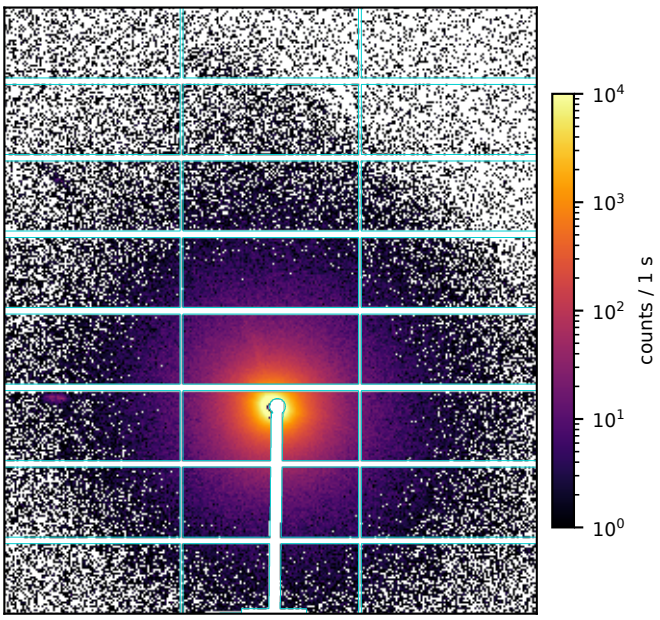
Detector and mask checks

Scan 1185490 | First / last accepted frames | Fixed scale within each detector

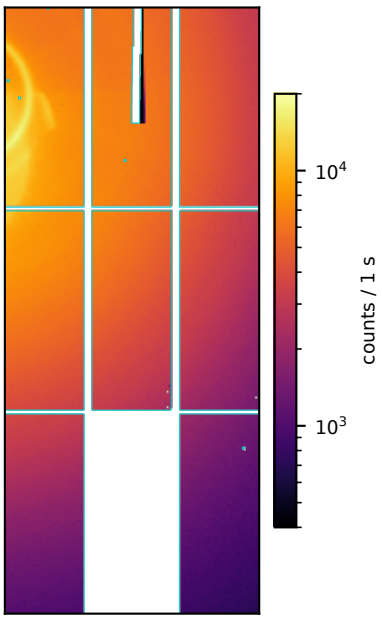
SAXS · event 1 · 16.33 h



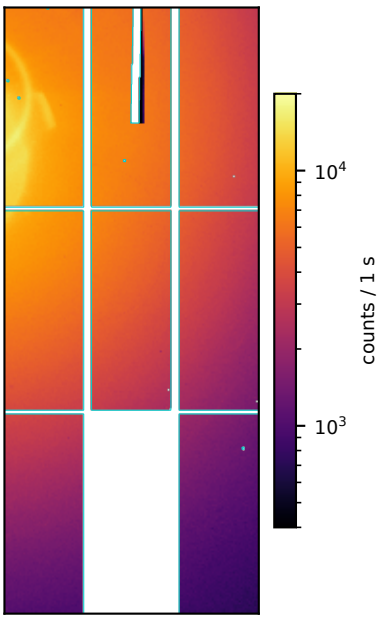
SAXS · event 113 · 25.62 h



WAXS · event 1 · 16.33 h



WAXS · event 113 · 25.62 h



SAXS: recorded pin beamstop, beam center (row, col) = (1104.30, 751.17) px; corrected SDD = 8.9863 m. Cyan outlines show excluded pixels. Negative Pilatus sentinels are rejected; zero counts remain valid for integration.
WAXS: 3-panel arc, 273 mm calibrated distance, arc approximately 15°. The saved bsx value is an overflow sentinel (−2147.483648 mm). The visible stationary shadow matches the unshifted bundled polygon; that fixed mask is used instead of applying a false motor-derived shift. The detector's bundled bad-module and gap masks are retained.
Post-trip events 28–35 of 1185487 retain substantial WAXS signal while SAXS is nearly dark. They are excluded by acquisition phase and are not used as dark-background references.

Methods, provenance and scan index

Reproducible reduction | All data access is read-only | Full frame/UID mapping in the CSV

REDUCTION

smi-tiled transmission integration: recorded 5999.9995 eV; flat Pilatus 2M SAXS geometry and folded 3-panel Pilatus 900KW WAXS geometry. SAXS uses recorded beam center and 9 m motor distance plus the package's calibration corrections; WAXS uses the package calibration, including −4.5 px column delta and +0.08 px/degree arc term. 2 × 2 pixel splitting; 700 SAXS / 1000 WAXS q bins; 180 azimuth bins; hot-pixel rejection threshold 3000, kernel 5. I(q) is a pixel-acceptance-weighted mean over the available azimuth, not a full-azimuth reconstruction. Package q values (nm^{-1}) are divided by 10 for $Å^{-1}$.

Normalize by recorded detector acquisition time (1 s), pixel solid angle and pre-trigger incident BPM3 (multiply by 50/BPM3). Do not normalize by the downstream diode because that would absorb sample-transmission changes into the scaling. Detector efficiencies are not cross-calibrated, so SAXS and WAXS are displayed independently. Small arc-readback variations are regridded linearly within each detector; missing coverage remains blank.

Comparisons are within sessions. Baseline sample-stage positions differ between October 22 and Saturday, and start/end piezo values vary during the long session. Per-frame sample positions were not recorded in primary. A scattering change therefore cannot be attributed uniquely to radiation chemistry or assigned a local dose from this record alone. Detector-dark rejection requires median WAXS = 0 and total nonnegative counts <10,000; raw totals and every exclusion are preserved in the audit CSV.

EXACT SCAN INDEX

WAXS-only, one image each:
1185420, 1185421, 1185422, 1185423, 1185424, 1185425, 1185426, 1185427
1185428, 1185429, 1185430, 1185431, 1185432, 1185433, 1185434, 1185435
1185436, 1185437, 1185438, 1185439, 1185440, 1185441, 1185442, 1185443
1185444, 1185445, 1185446, 1185447, 1185448, 1185449, 1185450, 1185451
1185452, 1185453, 1185454, 1185455, 1185456, 1185457, 1185458, 1185459
1185460, 1185461, 1185462, 1185463, 1185464, 1185465, 1185466, 1185467
1185468, 1185469, 1185470, 1185471, 1185472, 1185473, 1185474, 1185475
1185476, 1185477, 1185478, 1185479, 1185480, 1185481

SAXS + WAXS (1185489 is detector-dark):
1185487: fca35755-e42b-4d04-9d0b-8d2c57e96702
1185489: 2e6ebd11-15e9-4908-98e0-d24adab4fef2
1185490: 3b958206-68d7-4659-9dac-164624aa3911

REPRODUCTION AND FILES

From /home/xf12id/smi-tiled:
pixi run python ../SWAXS_user_scripts/Commissioning/2026-10-22_kapton_air_exposure/scattering_addendum.py --apply --download --reduce --report
Omit --download for offline reproduction. The addendum CSV identifies every acquired frame and its exposure; _iq.csv exports accepted profiles; logs/scattering_cache contains raw arrays, per-run reductions and geometry/mask hashes. The JSON records quantitative metrics and the immutable source database SHA-256. Analysis prepared with OpenCode (GPT-6 Astra).